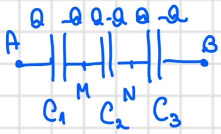
Gruparea în serie a condensatoarelor:



$C_{AB} = C_{es} = ?$

$$C = \frac{Q}{U_{12}} = \frac{Q}{V_1 - V_2}$$

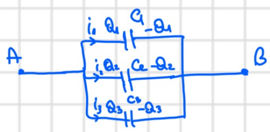
$$\left.\begin{array}{l} C_1 = C_{AM} = \dfrac{Q}{V_A - V_M} \Rightarrow V_A - V_M = \dfrac{Q}{C_1} \\[2ex] C_2 = C_{MN} = \dfrac{Q}{V_M - V_N} \Rightarrow V_M - V_N = \dfrac{Q}{C_2} \\[2ex] C_3 = C_{NB} = \dfrac{Q}{V_N - V_B} \Rightarrow V_N - V_B = \dfrac{Q}{C_3} \end{array}\right\} \xRightarrow{(+)} V_A - V_B = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3} = Q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)$$

$$\Updownarrow$$

$$C_{es} = \frac{Q}{V_A - V_B} = \frac{\cancel{Q}}{\cancel{Q}\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)} \Rightarrow C_{es} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}} \approx R_{ep}$$

$$C_{AB} = C_{es} = \frac{Q}{V_A - V_B}$$

Gruparea în paralel a condensatoarelor:



$C_{AB} = C_{ep} = ?$ $\left(C_{AB} = C_e = \dfrac{Q}{V_A - V_B}\right)$

$$\left.\begin{array}{l} C_1 = \dfrac{Q_1}{V_A - V_B} \Rightarrow Q_1 = C_1 (V_A - V_B) \\[2ex] C_2 = \dfrac{Q_2}{V_A - V_B} \Rightarrow Q_2 = C_2 (V_A - V_B) \\[2ex] C_3 = \dfrac{Q_3}{V_A - V_B} \Rightarrow Q_3 = C_3 (V_A - V_B) \end{array}\right\} \xRightarrow{(+)} Q_1 + Q_2 + Q_3 = (V_A - V_B)(C_1 + C_2 + C_3)$$

$$Q = Q_1 + Q_2 + Q_3$$

$$C_{ep} = \frac{Q}{V_A - V_B} = \frac{\cancel{(V_A - V_B)}(C_1 + C_2 + C_3)}{\cancel{(V_A - V_B)}} \Rightarrow C_{ep} = C_1 + C_2 + C_3$$

$$W_{el} = \frac{1}{2} \cdot \sum_{i=1}^{\infty} Q_i V_i$$

$$W_{el} = \begin{cases} \frac{1}{2} Q \cdot U_{AB} \\ \frac{1}{2} \cdot \frac{Q^2}{C} \\ \frac{1}{2} \cdot C U_{AB}^2 \end{cases}$$

Prob examen:

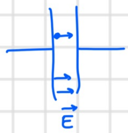
1. $C = 2\mu F = 2 \cdot 10^{-6} F$

$U = 50 V$

$W_{el} = ?$

$W_{el} = \frac{1}{2} \cdot C \cdot U^2 = \frac{1}{\cancel{2}} \cdot \cancel{2} \cdot 10^{-6} \cdot 2500 = 2{,}5 \cdot 10^{-3} J$

1. (Robotică)



$L = F \cdot d$

$\downarrow$

$q \cdot E$

$U = \frac{L}{q}$

$D = 3 \cdot 10^{-2} m$

$d = 2 \cdot 10^{-3} m$

$E = 10^5 V/m$

a) $U = ?$

b) $Q = ?$

$$\boxed{\begin{gathered} C = \frac{Q}{L} = \frac{\varepsilon_0 \cdot S}{d} \\ U = \frac{L}{q} \\ L = Fd = qE \end{gathered}}$$

$Fd = qE$

a) $U = \frac{L}{q} = \frac{\cancel{q}Ed}{\cancel{q}} = Ed = 10^5 \cdot 2 \cdot 10^{-3} = 2 \cdot 10^2 V$

b) $R = 1{,}5 \cdot 10^{-2} m$

$S = \pi R^2 = 225 \cdot 10^{-6} \pi \cdot m^2$

$C = \frac{8{,}85 \cdot 10^{-12} \cdot 225 \cdot \pi \cdot 10^{-6}}{2 \cdot 10^{-3}} = \ldots$

21/863

$C_1 = 6 \mu F$

$C_2 = 10 \mu F$

$C_3 = 16 \mu F$

$C_{ep} = ? = C_1 + C_2 + C_3 = 32 \cdot 10^{-6} F$

$C_{es} = ? = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}} = \frac{C_1 C_2 C_3}{C_2 C_3 + C_1 C_3 + C_1 C_2} = \frac{10 \cdot 16 \cdot 6}{160 + 96 + 60}$

$= \frac{10 \cdot 16 \cdot 6}{316} = \frac{240}{79} \mu F$

23/863

$C = 50\,\mu F$

$C_1 = 75\,\mu F$

$C_2 = ?$ e în paralel sau în serie ?

$50 = \dfrac{1}{\frac{1}{75} + \frac{1}{C_2}} \Rightarrow 50 = \dfrac{75C_2}{C_2 + 75} \Rightarrow 50C_2 + 3750 = 75C_2$

$\Rightarrow 25C_2 = 3750 \;|:5$

$\Rightarrow 5C_2 = 750$

$\Rightarrow C_2 = 150$

25...

27/863.

$C_2 = \frac{1}{2} C_1$

$U_2 = 2 \cdot U_1$

$\dfrac{W_{el_1}}{W_{el_2}} = ?$

$W_{el} = \begin{cases} \frac{1}{2} Q \cdot U_{AB} \\ \frac{1}{2} \cdot \frac{Q^2}{C} \\ \frac{1}{2} \cdot CU^2 \end{cases}$

$W_{el_1} = \frac{1}{2} \cdot C_1 \cdot \left(\frac{1}{2} U_2\right)^2$

$W_{el_2} = \frac{1}{2} \cdot \frac{1}{2} C_1 \cdot U_2^2$ (:)

$\dfrac{W_{el_1}}{W_{el_2}} = \dfrac{1}{2}$

29/863.

$D = 2 \cdot 10^{-2}\,m \Rightarrow R = 10^{-2}\,m \Rightarrow S = \pi \cdot 10^{-4}\,m^2$

$d = 5 \cdot 10^{-4}\,m$

$U = 200V$

a) $W_{el} = ?$

b) $\dfrac{W_{el}}{V} = ?$

$C = \dfrac{\varepsilon_0 S}{d} = \dfrac{8{,}85 \cdot 10^{-12} \cdot \pi \cdot \cancel{10^{-4}}}{5 \cdot \cancel{10^{-4}}} = 177\pi \cdot 10^{-14}\,F$

$W_{el} = \frac{1}{\cancel{2}} \cdot 4 \cdot 10^{4} \cdot 177 \cdot \pi \cdot 10^{-14} = 354\,\pi \cdot 10^{-10}\,J$